

Main Survey - correct

Survey Flow

Standard: Intro & Consent (1 Question)
Standard: Consent Statement: (2 Questions)
Standard: VVIQ (1 Question)

Branch: New Branch

If

If Consent Statement: By clicking "I agree", you confirm that: You are at least 18 years old You ha... No, I do not wish to participate Is Selected

EndSurvey: Advanced

Standard: VVIQ Block 1 (4 Questions)
Standard: VVIQ Block 2 (4 Questions)
Standard: VVIQ Block 3 (4 Questions)
Standard: VVIQ Block 4 (4 Questions)
Standard: Information (1 Question)

BlockRandomizer: 1 - Evenly Present Elements

EmbeddedData
target_word = Justice
EmbeddedData
target_word = Wisdom
EmbeddedData
target_word = Infinity
EmbeddedData
target_word = House
EmbeddedData
target_word = Glacier
EmbeddedData
target_word = Anger
EmbeddedData
target_word = Microscope
EmbeddedData
target_word = Laziness
EmbeddedData
target_word = Mirror
EmbeddedData
target_word = Lake
EmbeddedData
target_word = Night
EmbeddedData
target_word = Man
EmbeddedData
target_word = Nostalgia
EmbeddedData

target_word = Steering wheel

EmbeddedData

target_word = Euphoria

EmbeddedData

target_word = Wig

EmbeddedData

target_word = Anarchy

EmbeddedData

target_word = Margarita

EmbeddedData

target_word = Love

EmbeddedData

target_word = Luck

Standard: Experiment: Mental Image Task (30 Questions)

Standard: Experiment Result (5 Questions)

Standard: Overall Experience (3 Questions)

Standard: Demographics (3 Questions)

Page Break

Purpose of this study

You are invited to take part in a study examining how people visualize concepts in their mind and how generative AI tools (e.g., ChatGPT image generation) may help or influence this process. GenAI (Generative Artificial Intelligence) refers to a type of artificial intelligence that can create new content (text, music, videos, images, etc.), rather than only analyzing or classifying existing data. The study investigates how closely AI-generated images match a person's mental image and whether this interaction changes the mental image over time. What you will be asked to do if you agree to take part: You will imagine a word and briefly describe your mental images. You will generate images using the generative image tool ChatGPT image generator. You will evaluate how well these images match your mental image. You will answer questions about your experience and how your mental images may have changed. The study will take approximately **20–25 minutes** to complete. **For your convenience, it is best to use a laptop or desktop computer.** **Your Rights:**

Participation is voluntary. You may withdraw at any time without penalty. You may skip any question you do not wish to answer. No identifying information will be collected.

You may answer in German and/or English. **Confidentiality:** Your responses will be stored securely and analyzed anonymously. No IP addresses, names, or identifying information will be linked to your data. **Contact Information:** If you have questions about the study, please contact: Gwen Ghittini – gwen.ghittini@student.unisg.ch

End of Block: Intro & Consent

Start of Block: Consent Statement:

Q1 Consent Statement: By clicking “I agree”, you confirm that: You are at least 18 years old.
You have read and understood the information above. You voluntarily agree to
participate in this study. Do you agree to participate in this study?

- ☐ Yes, I agree to participate (1)
- ☐ No, I do not wish to participate (2)

Q2 You previously provided your email address to indicate your interest in participating in the self-experiment - thank you! To strengthen the analysis, we would like to link your responses from the pre-survey with the data from this self-experiment. **If you agree, please enter the same email address here so that your responses can be linked across surveys.** Your email address will be used exclusively for data linkage, stored separately from the survey responses, deleted after the data has been linked. Therefore the results will not be associated with you in any analyses or publications.

End of Block: Consent Statement:

Start of Block: VVIQ

Q3 How vivid is your mind's eye? The Vividness of Visual Imagery Questionnaire (VVIQ) explores the vividness of your visual imagination (Deutsch: Lebhaftigkeit der Vorstellung). It was created in 1973 by British psychologist David Marks and is proven to be an accurate test of the vividness with which you can see people, objects, or settings in your mind's eye. **How it works: Rate how vivid the image is by choosing one of the following answers:**

Perfectly clear and as vivid as normal vision

Clear and reasonably vivid

Moderately clear and vivid Vague and dim

No image at all, you only “know”

that you are thinking of an object In the next four question blocks, you will be asked to imagine the following: 1. a relative or friend you see often 2. a rising sun 3. the front of a familiar shop 4. a country scene After this test, the self-experiment will begin.

End of Block: VVIQ

Start of Block: VVIQ Block 1

Q4 Think of some relative or friend whom you frequently see (but who is not with you at present) and consider carefully the picture that comes before your mind's eye. Now imagine the exact contours of face, head, shoulders and body.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q5 Now imagine that persons characteristic poses of head, attitudes of body etc.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q6 Now imagine their precise carriage, length of step, etc., in walking.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q7 Imagine your relative or close friend wearing a familiar outfit in different colors.

- ☐ No image at all, you only know that you are thinking of the object (1)
- ☐ Vague and dim (4)
- ☐ Moderately clear and vivid (5)
- ☐ Clear and reasonably vivid (6)
- ☐ Perfectly clear and as vivid as normal vision (7)

End of Block: VVIQ Block 1

Start of Block: VVIQ Block 2

Q8 Visualize a rising sun. Consider carefully the picture that comes before your mind's eye. Now imagine the sun rising above the horizon into a hazy sky (dunstiger Himmel).

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q9 The sky clears and surrounds the sun with blueness.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q10 Imagine clouds gathering, and then a storm with flashes of lightning.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q11 A rainbow appears.

- ☐ No image at all, you only know that you are thinking of the object (1)
- ☐ Vague and dim (4)
- ☐ Moderately clear and vivid (5)
- ☐ Clear and reasonably vivid (6)
- ☐ Perfectly clear and as vivid as normal vision (7)

End of Block: VVIQ Block 2

Start of Block: VVIQ Block 3

Q12 Think of the front of a shop which you often go to. Consider the picture that comes before your mind's eye. Imagine the overall appearance of the shop from the opposite side of the road.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q13 A window display including colours, shapes and details of individual items.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q14 You are near the entrance: the colour, shape and details of the door.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q15 You enter the shop and go to the counter. The counter assistant serves you. Money changes hands.

- ☐ No image at all, you only know that you are thinking of the object (1)
- ☐ Vague and dim (4)
- ☐ Moderately clear and vivid (5)
- ☐ Clear and reasonably vivid (6)
- ☐ Perfectly clear and as vivid as normal vision (7)

End of Block: VVIQ Block 3

Start of Block: VVIQ Block 4

Q16 **Finally, think of a country scene which involves trees, mountains, and a lake. Consider the picture that comes before your mind's eye.** The contours of the landscape.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q17 The colour and shape of the trees.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q18 The colour and shape of the lake.

- ☐ No image at all, you only know that you are thinking of the object (1)
 - ☐ Vague and dim (4)
 - ☐ Moderately clear and vivid (5)
 - ☐ Clear and reasonably vivid (6)
 - ☐ Perfectly clear and as vivid as normal vision (7)
-

Q19 A strong wind blows on the trees and on the lake, causing waves.

- ☐ No image at all, you only know that you are thinking of the object (1)
- ☐ Vague and dim (4)
- ☐ Moderately clear and vivid (5)
- ☐ Clear and reasonably vivid (6)
- ☐ Perfectly clear and as vivid as normal vision (7)

End of Block: VVIQ Block 4

Start of Block: Information

Q20 Thank you for filling out the Vividness of Visual Imagery Questionnaire. Next you will conduct a self-experiment.

End of Block: Information

Start of Block: Experiment: Mental Image Task

Q21 In the following task, you will be asked to form a mental image for a word. Unlike the previous questions, some words may refer to abstract concepts (e.g., ideas, values, emotions). There is no right or wrong way to imagine them. People often visualize these words in different ways, such as a symbol, a scene, an object, a person, a place, an atmosphere, or even just colors and shapes. Please take a moment to form a mental image for the following word: **"\${e://Field/target_word}"** **Take your time (about 1–2 minutes)**. If it helps, notice what comes to mind first, then add detail. You might consider: what it looks like (colors, shapes, brightness) what it contains (objects, people, symbols) where it is (setting, background, scale, distance) what it feels like (mood, tension, calm, movement) Your image can be figurative or abstract, whatever feels natural.

	Not vivid at all (1)	Very weak (2)	Weak (3)	Moderately vivid (4)	Quite vivid (5)	Very vivid (6)	Extremely vivid (7)
How vivid is the mental image you are imagining right now? (1)	☐	☐	☐	☐	☐	☐	☐

Q22 Please briefly describe your mental image of **"\${e://Field/target_word}"** in 1–3 sentences. Describe it as accurately as possible (colors, components, scenes, other elements or the organization of the entire scene)

Page Break

Now please open ChatGPT in your browser. If you do not have an account, please sign up for the free version using your email address.

Page Break

Q23 In this part, you will try to create an image with a generative AI tool that matches your mental image of "**`{e://Field/target_word}`**" as closely as possible. You will use the generative image tool ChatGPT image generator. You will get 2 further prompts to try to get as close to your mental image as possible. For the next two rounds: Use your current mental image of "**`{e://Field/target_word}`**". Write a prompt in ChatGPT to generate your mental picture of the term given to you above. Generate an image. Download the generated image and upload it here. **Do not close the ChatGPT tab, as you will continue the conversation for the next 2 rounds.**

Q24 Please copy the prompt you used in the text box below.

Q25 Please download your generated image and upload it here (you can also drag and drop the file into this field).

Page Break

Round 1

Q26 **Image agreement (overall):** How closely does the image you just generated match what you were imagining *before* you saw it?

	Not at all (1)	Very weakly (2)	Weakly (3)	Moderately (4)	Strongly (5)	Very strongly (6)	Almost exactly (7)
1 = Not at all ... 7 = Almost exactly (1)	☐	☐	☐	☐	☐	☐	☐

Q27 Image agreement subscales: Please answer the following questions, whereby 1 = Strongly disagree ... 7 = Strongly agree

	Strongly disagree (1)	Disagree (2)	Slightly disagree (3)	Neither agree nor disagree (4)	Slightly agree (5)	Agree (6)	Strongly agree (7)
The scene or composition (arrangement of elements) matches my mental image. (1)	☐	☐	☐	☐	☐	☐	☐
The overall atmosphere or style matches my mental image. (2)	☐	☐	☐	☐	☐	☐	☐
Important details and components from my mental image are present in this image. (3)	☐	☐	☐	☐	☐	☐	☐

Q28 Change in Mental Image: Please indicate how much you agree with the following statement: "My mental image has changed compared to the mental image I had just before generating this AI-generated image."

	Strongly disagree (1)	disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
1 = Not changed at all ... 7 = Completely changed (1)	☐	☐	☐	☐	☐	☐	☐

Q29 If your mental image changed, what aspects changed? Please describe as accurately as possible.

- ☐ symbol (4)
 - ☐ scenes (5)
 - ☐ objects (6)
 - ☐ people (7)
 - ☐ places (8)
 - ☐ the atmosphere (9)
 - ☐ colors (10)
 - ☐ shapes (11)
 - ☐ nothing changed (13)
 - ☐ other (please describe shortly) (12)
-

Q30 Do you have any additional comments or thoughts you would like to share?

Page Break

Round 2

Q31 Assuming that there is still a mismatch between your mental image and your generated image: Your task now is to try to fill this gap and to add a prompt to your ChatGPT conversation to more accurately represent your mental image. Use your current mental image of **"\${e://Field/target_word}"**. Write a prompt in ChatGPT to represent your mental picture more accurately. Generate the new image. Download the generated image and upload it here.

Do not close the ChatGPT tab, as you will continue the conversation for the final round.

Q32 Please copy the prompt you used in the text box below.

Q33 Please download your generated image and upload it here (you can also drag and drop the file into this field).

Page Break

Q34 Image agreement (overall): How closely does the new/second image you just generated match what you were imagining *before* you saw it?

	Not at all (1)	Very weakly (2)	Weakly (3)	Moderately (4)	Strongly (5)	Very strongly (6)	Almost exactly (7)
1 = Not at all ... 7 = Almost exactly (1)	☐	☐	☐	☐	☐	☐	☐

Q35 Image agreement subscales: Please answer the following questions, whereby 1 = Strongly disagree ... 7 = Strongly agree

	Strongly disagree (1)	Disagree (2)	Slightly disagree (3)	Neither agree nor disagree (4)	Slightly agree (5)	Agree (6)	Strongly agree (7)
The scene or composition (arrangement of elements) matches my mental image. (1)	☐	☐	☐	☐	☐	☐	☐
The overall atmosphere or style matches my mental image. (2)	☐	☐	☐	☐	☐	☐	☐
Important details and components from my mental image are present in this image. (3)	☐	☐	☐	☐	☐	☐	☐

Q36 **Change in Mental Image:** Please indicate how much you agree with the following statement: "My mental image has changed compared to the mental image I had just before generating this AI-generated image."

	Strongly disagree (1)	disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
1 = Not changed at all ... 7 = Completely changed (1)	☐	☐	☐	☐	☐	☐	☐

Q37 If your mental image changed, what aspects changed? Please describe as accurately as possible.

Q38 Do you have any additional comments or thoughts you would like to share?

Page Break

Round 3

Q39 Assuming that there is still a mismatch, even a small one, between your mental image and your generated image after the second round: Even if your image is already quite close to what you imagine it to be, your task now is to try to fill the remaining gap further and to add one last prompt to your ChatGPT conversation to more accurately represent your mental image.

Use your current mental image of "[\\${e://Field/target_word}](#)". Write a prompt in ChatGPT to represent your mental picture more accurately. Generate the new image.

Download the new generated image and upload it here. **Do not close the ChatGPT tab, as you will continue the conversation for the final round.**

Q40 Please copy the prompt you used in the text box below.

Q41 Please download your generated image and upload it here (you can also drag and drop the file into this field).

Page Break

Q42 Image agreement (overall): How closely does the new/third image you just generated match what you were imagining before you saw it?

	Not at all (1)	Very weakly (2)	Weakly (3)	Moderately (4)	Strongly (5)	Very strongly (6)	Almost exactly (7)
1 = Not at all ... 7 = Almost exactly (1)	☐	☐	☐	☐	☐	☐	☐

Q43 Specific aspects of image agreement: Please answer the following questions, whereby 1 = Strongly disagree ... 7 = Strongly agree

	Strongly disagree (1)	Disagree (2)	Slightly disagree (3)	Neither agree nor disagree (4)	Slightly agree (5)	Agree (6)	Strongly agree (7)
The scene or composition (arrangement of elements) matches my mental image. (1)	☐	☐	☐	☐	☐	☐	☐
The overall atmosphere or style matches my mental image. (2)	☐	☐	☐	☐	☐	☐	☐
Important details and components from my mental image are present in this image. (3)	☐	☐	☐	☐	☐	☐	☐

Q44 **Change in Mental Image:** Please indicate how much you agree with the following statement: "My mental image has changed compared to the mental image I had just before generating this AI-generated image."

	Strongly disagree (1)	disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
1 = Not changed at all ... 7 = Completely changed (1)	☐	☐	☐	☐	☐	☐	☐



Q45 If your mental image changed, what aspects changed? Please describe as accurately as possible.

Q46 Do you have any additional comments or thoughts you would like to share?

End of Block: Experiment: Mental Image Task

Start of Block: Experiment Result



Q47 Please describe your current mental image of "**`{e://Field/target_word}`**".

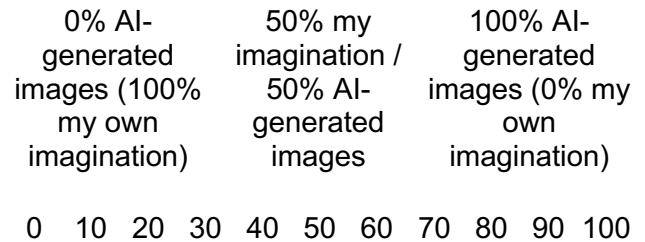
Q48 Similarity to original: How similar is your current mental image to your original one for "**`{e://Field/target_word}`**"?

	Completely different (1)	Very different (2)	Somewhat different (3)	Moderately similar (4)	Quite similar (5)	Very similar (6)	Essentially the same (7)
1 = Completely different ... 7 = Essentially the same (1)	☐	☐	☐	☐	☐	☐	☐

Q49 Source of current image: Which best describes your current mental image of "\${e://Field/target_word}"?

- ☐ Mostly my original imagination (1)
- ☐ A blend of my imagination and the AI images (2)
- ☐ Mostly based on the AI images (3)
- ☐ Not sure (4)

Q50 Please think about your current mental image of the word you worked on after completing all image-generation rounds. To what extent is this mental image based on your own original imagination versus the AI-generated images you saw during the task?

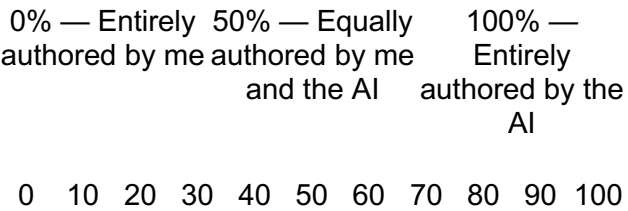


Please move the slider to indicate the relative contribution of your own imagination and the AI-generated images. (4)



Page Break

Q51 Please think about your current mental image of the word you worked on after completing all image-generation rounds. To what extent do you consider yourself versus the AI as the author of your current mental image?



Please move the slider to indicate the relative authorship of your mental image by you versus the AI. (4)	
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End of Block: Experiment Result

Start of Block: Overall Experience

Q52 Across the entire task, how strongly did the AI-generated images change your mental images?

	Not at all (2)	Very weakly (8)	Weakly (9)	Moderately (10)	Strongly (11)	Very strongly (12)	Extremely strongly (13)
1 = Not at all ... 7 = Extremely strongly (1)	☐	☐	☐	☐	☐	☐	☐

Q53 Across the entire task, did you generally prefer your own mental images or the AI-generated images?

	Strongly prefer my own mental images (1)	Prefer my own mental images (2)	Slightly prefer my own mental images (3)	No preference (4)	Slightly prefer AI- generated images (5)	Prefer AI- generated images (6)	Strongly prefer AI- generated images (7)
1 = Strongly prefer my own mental images...7 = Strongly prefer AI- generated images (1)	☐	☐	☐	☐	☐	☐	☐

Q54 When your mental image changed, what type of change happened most often?

- ☐ Small tweaks to details (1)
- ☐ Adding new elements inspired by the AI (2)
- ☐ Replacing major parts of my image with the AI's version (3)
- ☐ My mental images rarely changed (4)
- ☐ Other (please describe) (5)

End of Block: Overall Experience

Start of Block: Demographics

Q55 How old are you?

Q56 What is your gender?

- ☐ Male (1)
- ☐ Female (2)
- ☐ Non-binary / third gender (3)
- ☐ Prefer not to say (4)

Q57 If you are interested in your VVIQ (Vividness of Visual Imagery Questionnaire) score and learning more about the findings of the study, you can leave me your e-mail.

- ☐ No thanks. (1)
- ☐ Yes please. (please write your e-mail in the text box below) (2)

End of Block: Demographics
